

13. Sea $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ definida por:

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + 2y^2} & \text{si } (x, y) \neq (0, 0) \\ 0 & \text{si } (x, y) = (0, 0) \end{cases}$$

Analice si f es continua en $(0, 0)$, en base a su conclusión **opine** con fundamento si la función puede tener derivadas parciales continuas en un entorno de dicho punto.

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L \Rightarrow \lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L \quad \forall C \quad (x_0,y_0) \in C.$$

$$\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) \neq \exists \Rightarrow \lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = \cancel{\exists}$$

→ o vale distinto según C .

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + 2y^2} & \text{si } (x, y) \neq (0, 0) \\ 0 & \text{si } (x, y) = (0, 0) \end{cases}$$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y)$$

$$\lim_{\substack{x \rightarrow 0 \\ y = mx}} \frac{x \cdot mx}{x^2 + 2m^2x^2} = \frac{m}{1+2m^2} \rightarrow \cancel{\lim} \Rightarrow \text{No es cont.}$$

$$f \text{ dif} \Rightarrow \begin{cases} f \text{ cont} \\ f \text{ deriv.} \\ f'(A, \vec{r}) = \nabla f(A) \cdot \vec{r} \end{cases}$$

$$f \in C^1 \Rightarrow f \text{ dif.}$$

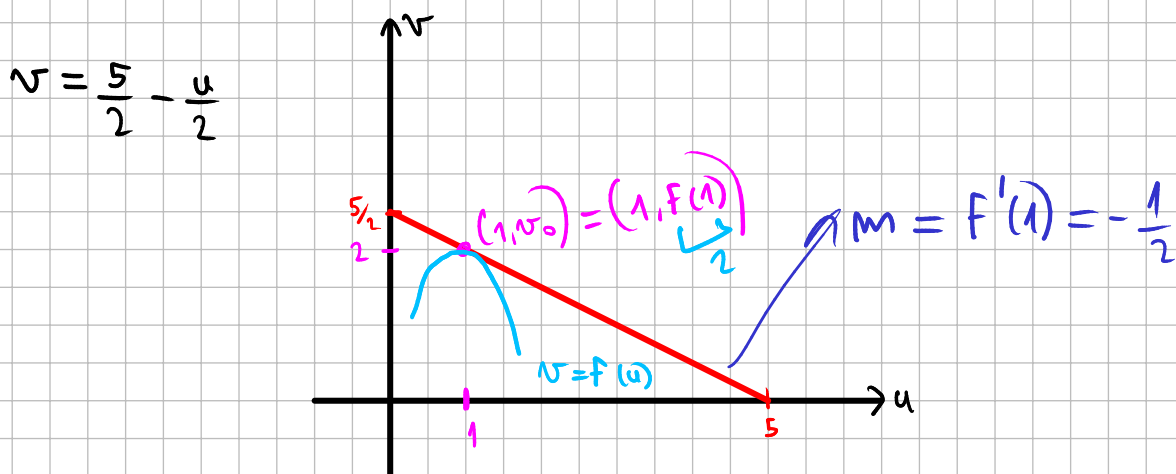
$$f \text{ No es dif} \Rightarrow f \text{ No es } C^1$$

$$f \text{ No es cont} \Rightarrow f \text{ No es dif} \Rightarrow f \text{ No es } C^1$$

14. Dada $h(x, y) = f(x^2 - y)$ con $f \in C^1(\mathbb{R})$ tal que la curva de ecuación $v = f(u)$ admite recta tangente $u + 2v = 5$ en el punto $(1, v_0)$, **calcule** una aproximación lineal para $h(0.9, 0.2)$.

$$h(x, y) \approx h(1, 0) + h'_x(1, 0)(x-1) + h'_y(1, 0) \cdot y \quad \text{en } E(1, 0)$$

$$h(1, 0) = f(1) = 2$$



$$h(x, y) = f(\underbrace{x^2 - y})$$

Dom.

$\mathbb{R}$ Func

$\mathbb{R}^n$ CAMPO

Esc

VECT.

$\mathbb{I}_m$

$\mathbb{R}^m$

$$g(x, y) : \text{C.E.}$$

$$f(u) : \text{F.E.}$$

$$h(x, y) : \text{C.E.}$$

$$h = f \circ g \Rightarrow h \text{ dif.}$$

$$Dh(1, 0) = Df(g(1, 0)) \cdot Dg(1, 0)$$

$$Dh(1, 0) = Df(1) \cdot Dg(1, 0)$$

$$Df(1) = f'(1) = -\frac{1}{2}$$

$$g(x, y) = x^2 - y$$

$$Dh(1, 0) = -\frac{1}{2} \cdot [2 \ -1]$$

$$Dg(x, y) = [2x \ -1]$$

$$Dh(1, 0) = [-1 \ \frac{1}{2}]$$

$$h(x, y) \approx 2 - (x-1) + \frac{1}{2} \cdot y$$

$$h(0.9, 0.2) \approx 2 - (0.9-1) + \frac{1}{2} \cdot 0.2 = 2.2$$

$$h(0.9, 0.2) \approx 2.2$$

1. Sea $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ definida por:

$$f(x, y) = \begin{cases} \frac{x^2 \sin(y)}{x^2 + y^2} & \text{si } (x, y) \neq (0, 0) \\ 0 & \text{si } (x, y) = (0, 0) \end{cases}$$

~~Analice la derivabilidad de f según distintas direcciones en el punto $(0, 0)$.~~

f dif?

fuera del 0,0 f es diferenciable por ser división de funciones diferenciables con denominador no nulo.

$$f \text{ dif} \Rightarrow \begin{cases} f \text{ cont.} \\ f \text{ deriv.} \\ f'(A, \vec{r}) = \nabla f(A) \cdot \vec{r} \quad \forall (a, b) \end{cases}$$

$$f(x, y) = \begin{cases} \frac{x^2 \sin(y)}{x^2 + y^2} & \text{si } (x, y) \neq (0, 0) \\ 0 & \text{si } (x, y) = (0, 0) \end{cases}$$

$$\lim_{x \rightarrow 0} \frac{x^2 \cdot \sin(mx)}{x^2 + m^2 x^2} = \lim_{x \rightarrow 0} \frac{\sin(mx)}{1 + m^2} = 0$$

$$0 \leq x^2 \leq x^2 + y^2$$

$$0 \leq \frac{x^2}{x^2 + y^2} \leq 1$$

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 \cdot \sin y}{x^2 + y^2} = 0$$

$$f(0, 0) = 0$$

$\rightarrow f$ cont. en el $(0, 0)$.

$$f'(A, \vec{r}) = \lim_{h \rightarrow 0} \frac{f(A+h\vec{r}) - \overbrace{f(A)}^{=0}}{h}$$

$$A = (0,0)$$

$$\vec{r} = (a,b)$$

$$a^2 + b^2 = 1$$

$$f'((0,0), \vec{r}) = \lim_{h \rightarrow 0} \frac{f(ha, hb)}{h}$$

$$f(x,y) = \begin{cases} \frac{x^2 \sin(y)}{x^2 + y^2} & \text{si } (x,y) \neq (0,0) \\ 0 & \text{si } (x,y) = (0,0) \end{cases}$$

$$h \neq 0$$

$$(a,b) \neq (0,0)$$

$$\Rightarrow (ha, hb) \neq (0,0)$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h^2} \cdot a^2 \cdot \sin(hb)}{\underbrace{\cancel{h^2} a^2 + \cancel{h^2} b^2}_{=1}} \cdot \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a^2 \cdot \sin(hb)}{h} \stackrel{L'H}{=} \lim_{h \rightarrow 0} a^2 \cdot b \cdot \cos(hb) = \boxed{a^2 \cdot b}$$

$\Rightarrow f$ deriv en el origen.

$$f'_x(0,0) = 0$$

$$\nabla f(0,0) = (0,0)$$

$$f'_y(0,0) = 0$$

$$\nabla f(0,0) \cdot (a,b) = 0$$

$\neq$

$\Rightarrow$ No es dif.

$$\nabla f(0,0) = (c_1, c_2)$$

$$\nabla f \cdot \vec{r} = c_1 \cdot a + c_2 \cdot b$$

3. Siendo $h(x, y) = f(x - y^2, 2x + y)$ calcule aproximadamente $h(1.02, 0.99)$, usando aproximación lineal, sabiendo que $w = f(u, v)$ queda definida implícitamente mediante la ecuación $wu + \ln(w + v - u) = 0$ en un entorno de $(u_0, v_0) = (0, 3)$.

$$h(x, y) \approx h(1, 1) + h'_x(1, 1)(x - 1) + h'_y(1, 1)(y - 1)$$

$$h(1, 1) = f(0, 3)$$

$$h_w(w_0 + 3) = 0$$

$$w_0 = -2 = f(0, 3)$$

$$h(x, y) = f(\underbrace{x - y^2, 2x + y}_{\vec{g}(x, y)})$$

$$Dh(1, 1) = DF(\vec{g}(1, 1)) \cdot D\vec{g}(1, 1)$$

$$Dh(1, 1) = DF(0, 3) \cdot D\vec{g}(1, 1)$$

$$F(0, 3, -2) = 0$$

$$DF(u, v, w) \text{ cont en } E(0, 3, -2)$$

$$F'_w(0, 3, -2) \neq 0$$

Condiciones
(no necesario en este enunciado)

$$f'_u(0, 3) = - \frac{F'_u(0, 3, -2)}{F'_w(0, 3, -2)}$$

$$f'_v(0, 3) = - \frac{F'_v(0, 3, -2)}{F'_w(0, 3, -2)}$$

$$f'_u(0,3) = - \frac{F'_u(0,3,-1)}{F'_w(0,3,-2)}$$

$$\underbrace{wu + \ln(w + v - u)}_{F(u,v,w)} = 0$$

$$f'_v(0,3) = - \frac{F'_v(0,3,-1)}{F'_w(0,3,-2)}$$

$$F'_u = w - \frac{1}{w+v-u} \overset{(0,3,-2)}{=} -3$$

$$F'_v = \frac{1}{w+v-u} = 1$$

$$F'_w = u + \frac{1}{w+v-u} = 1$$

$$f'_u(0,3) = - \frac{F'_u(0,3,-1)}{F'_w(0,3,-2)} = 3$$

$$Df(0,3) = \begin{bmatrix} 3 & -1 \end{bmatrix}$$

$$f'_v(0,3) = - \frac{F'_v(0,3,-1)}{F'_w(0,3,-2)} = -1$$

$$\vec{f}(x, y) = (x - y^2, 2x + y)$$

$$D\vec{f} = \begin{bmatrix} 1 & -2y \\ 2 & 1 \end{bmatrix}$$

$$D\vec{f}(1, 1) = \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

$$Dh(1, 1) = [3 \quad -1] \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$$

$$Dh(1, 1) = [1 \quad -7]$$

$$h(x, y) \approx h(1, 1) + h'_x(1, 1)(x-1) + h'_y(1, 1)(y-1)$$

$$h(1.02, 0.99) \approx -2 + (1.02-1) - 7 \cdot (0.99-1)$$

$$\approx -1.91$$

24. Sea $C \subset \mathbb{R}^3$ la curva incluida en la superficie Σ de ecuación $z = xy + 4$ con $(x, y) \in \mathbb{R}^2$. Sabiendo que una ecuación vectorial para C es $\vec{X} = (t-1, t+1, g(t))$ con $t \in \mathbb{R}$, halle una ecuación para el plano normal a C en $(2, y_0, z_0)$.

$$(x, y, z) \cdot \vec{N} = \vec{N} \cdot A$$

$$(t_0-1, t_0+1, g(t_0)) = (2, y_0, z_0)$$

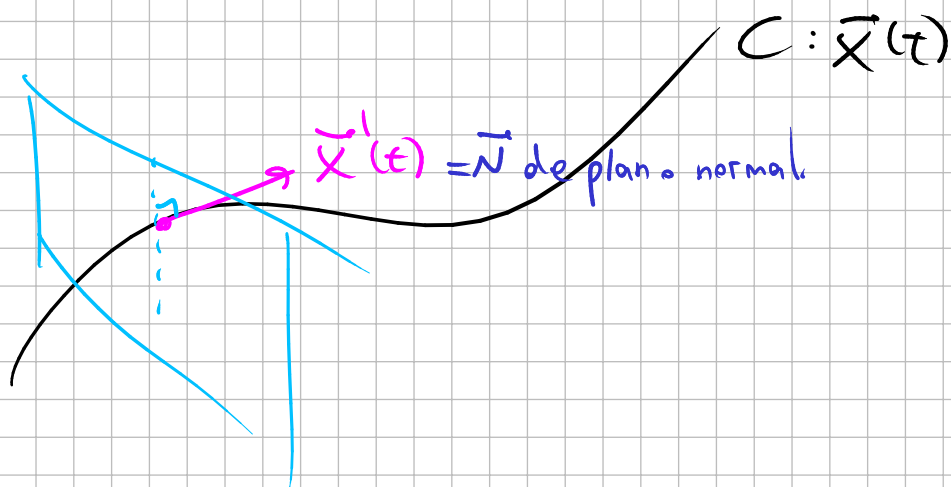
$$t_0 = 3$$

$$y_0 = 4$$

$$g(3) = z_0$$

$$z_0 = 2 \cdot 4 + 4 = 12$$

$$A = (2, 4, 12)$$



e Σ de ecuación $z = xy + 4$
 $\vec{X} = (t-1, t+1, g(t))$ con t

$$z = (t-1)(t+1) + 4 = t^2 - 1 + 4 = t^2 + 3 = g(t)$$

$$\vec{X} = (t-1, t+1, t^2+3)$$

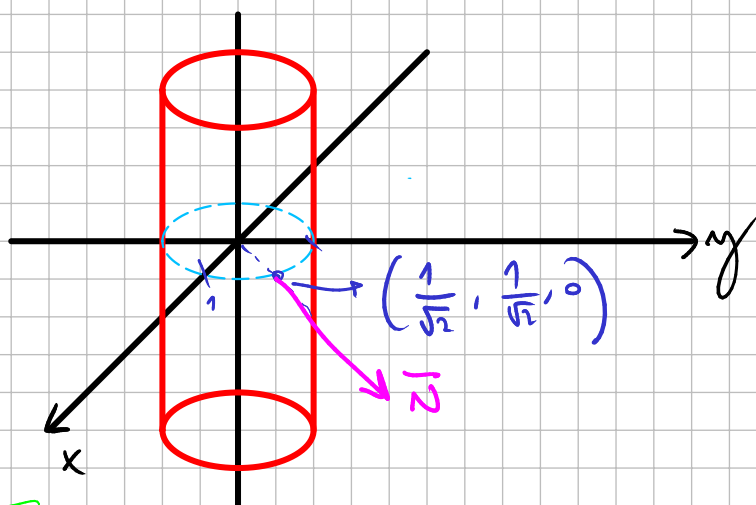
$$\vec{X}' = (1, 1, 2t)$$

$$\vec{X}'(3) = (1, 1, 6) = \vec{N}_\perp$$

$$(x, y, z) \cdot (1, 1, 6) = (1, 1, 6) \cdot (2, 4, 12)$$

$$x + y + 6z = 78$$

$$x^2 + y^2 = 1$$



$$F = K$$

$$\nabla F \perp F = K$$

$$F = x^2 + y^2$$

$$\nabla F = (2x, 2y, 0) = \vec{N}$$

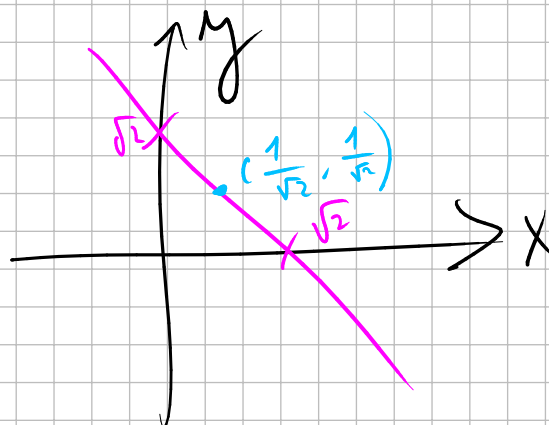
$$\vec{N}\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right) = (\sqrt{2}, \sqrt{2}, 0)$$

↳ uso $(1, 1, 0)$ (xq pintó :))

$$(1, 1, 0) \cdot (x, y, z) = (1, 1, 0) \cdot \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$$

$$x + y = \sqrt{2}$$

$$y = -x + \sqrt{2}$$



3. Dada $f: \mathbb{R}^2 \rightarrow \mathbb{R} / f(x,y) = 2x^2y - y^2$, **determine y grafique** el conjunto H del plano xy en cuyos puntos la función $h = f'_x f'_y$ resulta con valores positivos y **analice** si H es conexo.

$$f'_x = 4xy$$

$$f'_y = 2x^2 - 2y$$

$$H: 4xy(2x^2 - 2y) > 0$$

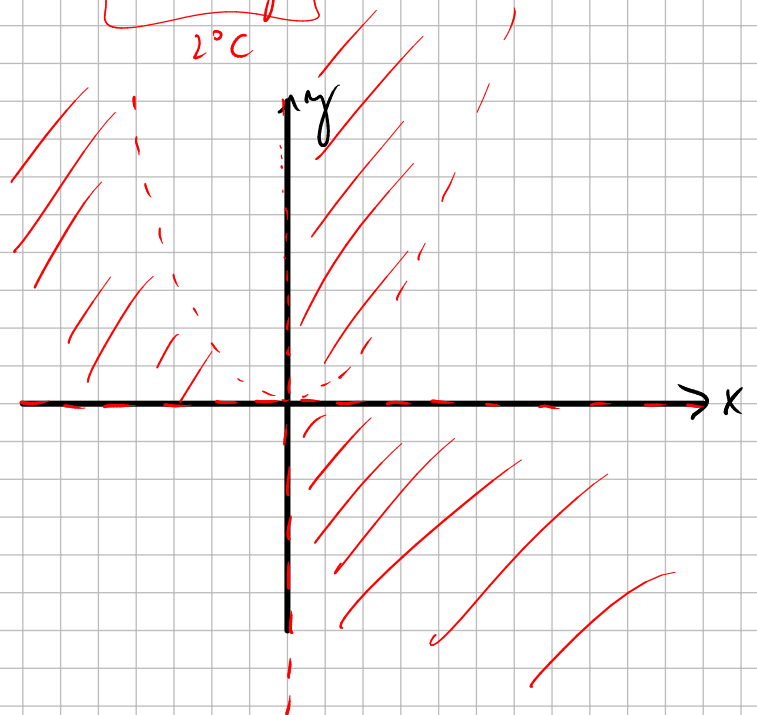
$$\underbrace{4xy > 0 \cap 2x^2 - 2y > 0}_{y > x^2}$$

$$\cup \underbrace{4xy < 0 \cap 2x^2 - 2y < 0}_{\substack{x > 0 \cap y < 0 \rightarrow 4^o C. \\ \cup x < 0 \cap y > 0 \\ 2^o C.}}$$

$$\underbrace{x > 0 \cap y > 0}_{1^o C.} \cup \underbrace{x < 0 \cap y < 0}_{3^o C.}$$

$$2x^2 - 2y = 0$$

$$y = x^2$$

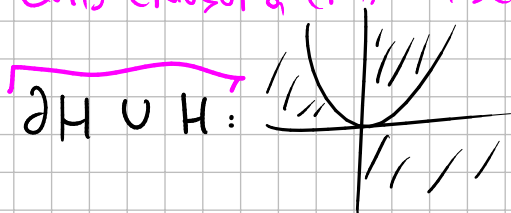


Abierto ✓
Cerrado ✗
Acotado ✗
Compacto ✗
Conexo ✗

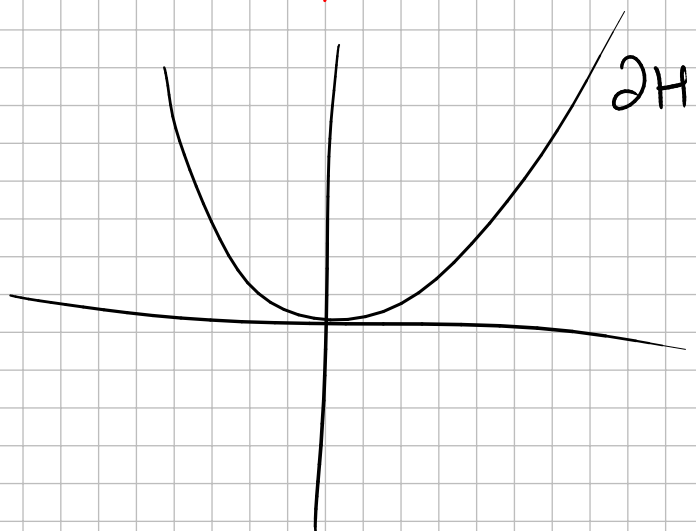
∂H

Punto interior: $(1, -1)$
Punto exterior: $(-1, -1)$
Punto de acum: $(0, 0), (1, -1)$
Punto frontera: $(0, 0), (0, 2)$

Catij Clausura (Pasa a ser cerrado)



$$\partial H \cap H: \emptyset$$



$$36x^2 > (6y-6)^2$$

$$\sqrt{36x^2} = \sqrt{(6y-6)^2}$$

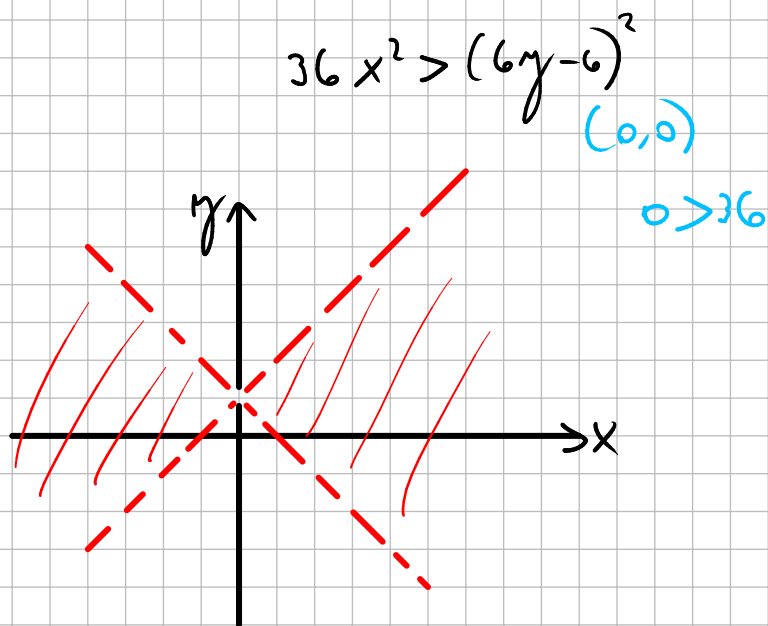
$$6|x| = |6y-6|$$

$$6|x| = 6|y-1|$$

$$|x| = |y-1|$$

$$\pm x = y-1$$

$$y = 1 \pm x$$



$$(3x^2 - 6xy + 3y^2, -6x + 6xy) = (0, 0)$$

$$-6x + 6xy = 0$$

$$x=0$$

$$-6y + 3y^2 = 0$$

$$3y(-2+y) = 0 \rightarrow y=2$$

$$\rightarrow y=0$$

$$xy = x \rightarrow x=0$$

$$\rightarrow y=1$$

$$P.C: (0,0) (0,2)$$

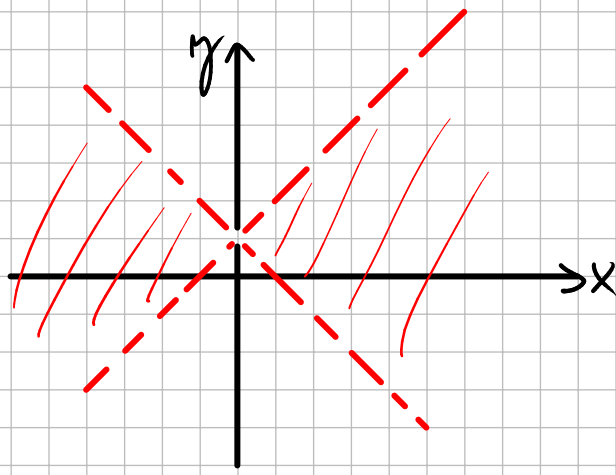
$$(1,1) (-1,1)$$

$$y=1$$

$$3x^2 - 6 + 3 = 0$$

$$3x^2 - 3 = 0 \rightarrow x = \pm 1$$

P.C: $(0,0)$ $(0,2)$
 $(1,1)$ $(-1,1)$



$$Hf = \begin{bmatrix} 6x & 6y-6 \\ 6y-6 & 6x \end{bmatrix}$$

$$Hf(1,1) = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix} \quad \begin{array}{l} H_1 = 6 > 0 \\ H_2 = 36 > 0 \end{array} \rightarrow \text{MIN. rel.}$$

$$H(-1,1) = \begin{bmatrix} -6 & 0 \\ 0 & -6 \end{bmatrix} \quad \begin{array}{l} H_1 = -6 \\ H_2 = 36 \end{array} \rightarrow \text{MAX. rel.}$$